

Progress Report

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Research Objective and Goal

The final goal of the project is to implement the distributed algorithm for PCA (Principal Component Analysis) developed by our lab and test its validity based on numerical experiment.

Short-Term Goals

10. Continue studying the number of iteration T_Y and T_Z in the averaging consensus of the matrices $Y_m(t)$ and $Z_m(t)$ and the number of iteration T_{PM} in the distributed algorithm for PCA alongside with star graphs, path graphs, circle graphs, and regular graphs.

Progress

In the last meeting, tables of different amount of iterations needed depending of the tolerance value of the convergence to the optimal value (CONSENSUS_EPS and CONVERGENCE_EPS) were presented.

As requested in the last meeting, here is a table showing the different amount of iterations needed for the decentralized algorithm for PCA to converge to the optimal with a fixed tolerance of error depending on the maximum number of iteration of the inner loop (T_Y and T_Z).

Here are the global variables:

SEED = 15

CONSENSUS_EPS = 1e-8

CONVERGENCE_EPS = 1e-10

T_{PM} = 10000

We can observe that for $T_Z \leq 14$, the algorithm does not converge to the optimal solution for the fixed tolerance error values within 10000 iterations of the outer loop.

We can also observe that the (global?) optimal overall number of iteration of the algorithm is reached at $42000 = 2000 \cdot (1 + 4 + 16)$, with 2000 iterations needed for the update rule, 4 iterations needed for the averaging consensus of Y and 16 iterations needed for the averaging consensus of Z.